Homework #1

AI Programming   Due: Oct. 3rd, 2021 23:59 (Sun)

Note: For the following exercises, write a program to solve the problem and display the answer. A possible output is shown in a shaded box and responses to input statements appear underlined.

Exercises 2.2

110. Word Replacement Write a program that requests a sentence, a word in the sentence, and another word and then displays the sentence with the first word replaced by the second. Do not use the format method.

```
Enter a sentence: What you don't know won't hurt you.
Enter word to replace: know
Enter replacement word: owe
What you don't owe won't hurt you.
```

Exercises 3.3

16. Bouncing Ball The coefficient of restitution of a ball, a number between 0 and 1, specifies how much energy is conserved when the ball hits a rigid surface. A coefficient of .9, for instance, means a bouncing ball will rise to 90% of its previous height after each bounce. Write a program to input a coefficient of restitution and an initial height in meters, and report how many times a ball bounces when dropped from its initial height before it rises to a height of less than 10 centimeters. Also report the total distance traveled by the ball before this point. The coefficients of restitution of a tennis ball, basketball, super ball, and softball are .7, .75, .9, and .3, respectively.

```
Enter coefficient of restitution: .7
Enter initial height in meters: 8
Number of bounces: 13
Meters traveled: 44.82
```

Exercises 3.4

74. A Puzzle The following puzzle is known as *The Big Cross-Out Swindle*. "Beginning with the word 'NAISNIENLGELTETWEORRSD,' cross out nine letters in such a way that the remaining letters spell a single word". Write a program that creates variables named *startingWord*, *crossedOutLetters*, and *remainingLetters*. The program should assign to *startingWord* the string given in the puzzle, assign to *crossedOutLetters* a list containing every other letter of *startingWord* beginning with the initial letter *N*, and assign to *remainingLetters* a list containing every other letter of *startingWord* beginning with the second letter, *A*. The program should then display the values of the three variables.

```
Starting word: NAISNIENLGELTETWEORRSD
Crossed out letters: N I N E L E T T E R S
Remaining letters: A S I N G L E W O R D
```

Exercises 4.1

26. Count Function Suppose the count function for a string didn't exist. Define a function that returns the number of non-overlapping occurrences of a substring in a string.

30. Pay Raise Write a pay-raise program that requests a person's first name, last name, and current annual salary, and then displays the person's salary for next year. People earning less than $40,000 will receive a 5% raise, and those earning $40,000 or more will receive a raise of $2,000 plus 2% of the amount over $40,000. Use functions for input and output, and a function to calculate the new salary.

```
Enter first name: John
Enter last name: Doe
Enter current salary: 48000
New salary for John Doe: $50,160.00
```